

VIBRATION SENSOR

ST IIS3DWB
Industrial
Accelerometer

- ODR: 26.67 kHz
 - Decimate /2
- > 13.33 / 12 kHz
- 3-Axis 16-bit
- 75 ug/v/Hz Noise
- SPI @ 6.0 MHz

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GOWIN KIWI 4K FPGA (PRIMARY TARGET: GW1NSR-LV4C) / PRIMER 20K (SCALABILITY)
[1000 MHz Core Clock] | [1000 MHz Streaming Clock] | [1000 MHz Core Clock]

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graph LR; S1["Sensor Interface & Decimator<br/>Polyphase FIR /2<br/>26.67 -> 13.33 kHz<br/>SPI Master 6 MHz"] --> S2["Ping-Pong<br/>BRAM Buffer<br/>2 x 512 x 16-bit<br/>Zero Sample Drop<br/>Double Buffering"]; S2 --> S3["Folded FIR Filter<br/>32-tap Min-Phase<br/>1 Multiplier @ 50MHz<br/>Group Delay: 1.33 ms"]; S2 --> S4["IIR Demodulator<br/>Full-Wave Rectifier<br/>DC Notch Filter<br/>0 CORDIC (Shift-Add)"]; S3 --> S5["512-pt Radix-2 FFT<br/>Gowin FFT IP Core<br/>Alpha-Beta Mag Engine<br/>257 Spectral Bins"]; S4 --> S6["Kinematic Prior<br/>fr Peak Detector<br/>(10 - 45 Hz Scan)<br/>4-bit Mask Vector"]; S5 --> S7["Deterministic<br/>Frame Assembler<br/><br/>CDC Async FIFO<br/>520B Frame + CRC16<br/>Assert FRAME_READY"]; style S3 fill:#d9e1f2,stroke:#333,stroke-width:1px; style S4 fill:#d9e1f2,stroke:#333,stroke-width:1px; style S5 fill:#d9e1f2,stroke:#333,stroke-width:1px; style S6 fill:#d9e1f2,stroke:#333,stroke-width:1px; style S7 fill:#d9e1f2,stroke:#333,stroke-width:1px;
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Kiwi 4K Budget: 1,650/4,608 LUT4 (35.8%) | 5/10 BSRAM (50%) | 6/16 DSP (37.5%) | Power: ~38 mW

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INTER-CHIP SPI SLAVE DMA BUS (4.0 MHz) + FRAME_READY IRQ
 512-Bit 520-Byte Frame: Sync (4B) | Cnt (2B) | Mask (2B) | 257 Bins (514B) | CRC16 (2B) |

SONIX SN32F788 MCU (ARM CORTEX-M4F @ 96 MHz, CPU, 256KB Flash, 32KB SRAM)
 3.3V Supply | 96 MHz CPU | 256KB Flash | 32KB SRAM | Sleep Efficiency: 67.4% | I_avg < 9.3 mA

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graph LR
    Input[Circular DMA Ring Buffer  
Direct to SRAM  
Hardware CRC-16  
Integrity Validated] --> PFFL[Physics-Feature Fusion Layer  
Kinematic Prior ⊗  
Multi-Scale Features]
    Input --> VDM[VibraDistillMicro 1D-CNN (INT8)  
  
8,677 Params (8.5KB)  
k=3,7,15 Depthwise  
CMSIS-NN: 12.88 ms]
    PFFL --> EDL[Evidential Dirichlet (EDL) Classifier Head  
  
Dirichlet α = e + 1  
Vacuity u = K/S (OOD Alert)]
    VDM --> BDA[Bounded-Delay ACP Engine Adaptive Conformal RUL  
  
[RUL_min, RUL_max]  
Certified 90% Coverage]
    EDL --> DPM[Dynamic Power Manager & Telemetry Pack  
  
13.9 ms Active / 28.8 ms Sleep  
WFI Deep Sleep (67.4%)  
Avg Current: 9.3 mA  
BOM < $15 USD]
    BDA --> DPM
  
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Decoupled Knowledge Distillation (DKD): ResNet Teacher (20.4k) -> VibraDistillMicro Student (8,677) [$\tau=4.0$, $\lambda_1=1.0$, $\lambda_2=1.0$]

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graph TD
    Top["0.96\" I2C OLED (SSD1306)  
• Fault: BPFO/BPFI  
• Vacuity u: 0.12 (OK)  
• RUL: [142, 168] h  
• Speed: fr = 29.8 Hz"]
    Bottom["Wireless / Fieldbus Telemetry  
• LoRaWAN (868/915) Long-range Smart Pod  
• RS485 Modbus RTU  
• CAN Bus 2.0B  
Industrial SCADA / PLC"]
    Top --> Bottom
    kHz[kHz] --> Top
    T_CAN[T/CAN] --> Bottom
  
```

FIELD TELEMETRY & INDUSTRIAL HMI

0.96" I2C OLED (SSD1306)

- Fault: BPFO/BPFI
- Vacuity u: 0.12 (OK)
- RUL: [142, 168] h
- Speed: fr = 29.8 Hz

kHz

Wireless / Fieldbus Telemetry

- LoRaWAN (868/915) Long-range Smart Pod
- RS485 Modbus RTU
 - CAN Bus 2.0B

Industrial SCADA / PLC

T/CAN

- **0.96" I2C OLED (SSD1306)**
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Telemetry**

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Long-range Smart Pod
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